Cosmin Coldea

Software Engineering master's student at the Technical University of Cluj-Napoca. I build full-stack apps and connected systems, and the part I care about most is **the interface people use to understand and control them**: dashboards for live device data, role-based tools, and clear flows for everyday tasks.

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TOOLBOX

Technical skills

FRONTEND

React, Angular, TypeScript, JavaScript, HTML, CSS, Recharts, Chart.js

BACKEND

Java, Spring Boot, REST APIs, WebSockets, RabbitMQ

LANGUAGES

Java, C/C++, Python

DATA & DEVOPS

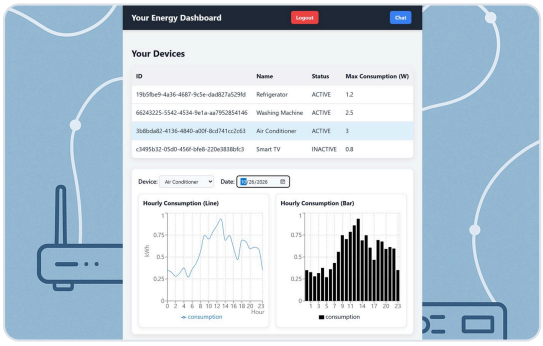
PostgreSQL, MySQL, MongoDB, Docker, Git, GitHub, Linux

ROBOTICS & VISION

ROS, Jetson Nano, Arduino, LiDAR, IMU, OpenCV, YOLO

SELECTED PROJECTS

Interfaces I designed and built



Client dashboard: devices and hourly consumption for a selected day

Smart Energy Management System

NOV 2025 – JAN 2026 SOLO PROJECT

A platform for monitoring the electricity consumption of smart devices. Clients follow their own devices, admins manage users and hardware, and overconsumption alerts arrive in real time.

DESIGN FOCUS

Making live device data readable for a non-technical client, and giving support a clear path from chatbot to a human.

- Client dashboard: pick a device and a day, then see the hourly consumption as both a line and a bar chart.
- Separate client and admin experiences, each showing only what that role needs.
- Support chat that answers common questions automatically (rules or AI) and hands over to an admin when needed.
- Real-time overconsumption notifications over WebSockets.

`React` `Recharts` `WebSockets` `Spring Boot` `RabbitMQ` `Docker`

GitHub repository ↗

TravelPoints

MAR – MAY 2026 TEAM OF 6

AGILE, 2-WEEK SPRINTS

A platform for discovering tourist attractions: browse and filter places, save favourites, leave reviews and plan trips. I worked across frontend and backend. The parts below are mine.

DESIGN FOCUS

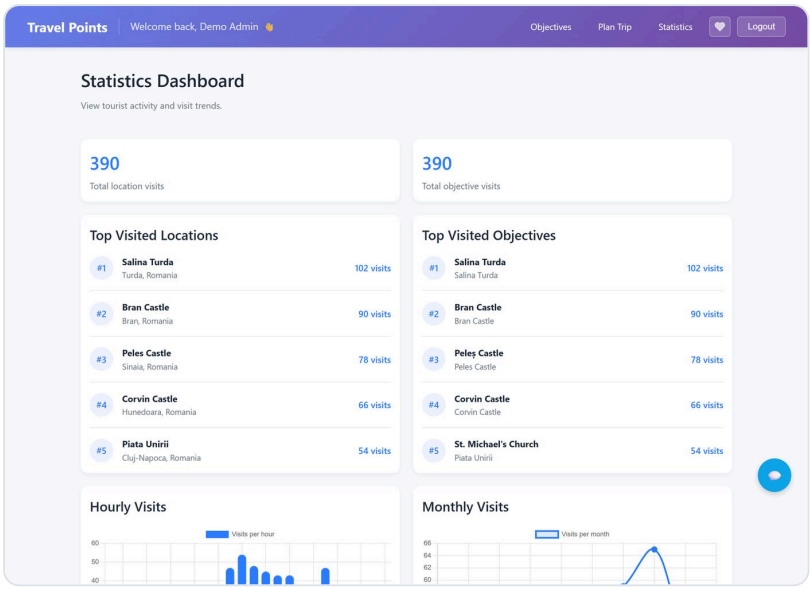
Turning raw visit data into a dashboard an admin can read at a glance, and making location search feel instant.

- Statistics dashboard laid out as summary first, detail after: KPI cards, then top-5 rankings, then hourly and monthly trend charts.
- "Near me" search. I measured where the delay came from (the browser's geolocation, not the database), then added location preloading and caching so repeat searches are instant.
- Login, registration and password-reset flows. The reset email now sends a direct link instead of a code to copy.
- Add-objective form with validation.

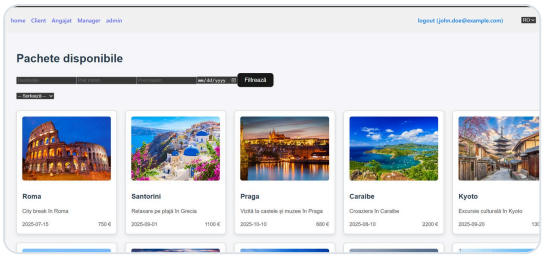
`Angular` `TypeScript` `Chart.js` `Spring Boot` `MongoDB geo queries`

Project showcase ↗

The team's source code is in a private university repository. The showcase covers my contributions.



Admin statistics dashboard



Package catalogue with filters, sorting and language switcher

Travel Agency Platform

SPRING 2025 SOLO PROJECT

A booking platform for a travel agency, used by four kinds of people: clients, employees, managers and admins.

DESIGN FOCUS

One product, four roles: each role gets its own screens and only the actions it needs.

- Visual catalogue of holiday packages with filtering by destination, price and date, plus sorting.
- Role-based navigation and dashboards for bookings, clients, users and sales statistics.
- Interface fully translated into Romanian, English and French, switchable at runtime.
- Export of packages, users and bookings to CSV, JSON, XML or Word.

`React` `i18next` `Spring Boot` `Microservices` `MySQL`

GitHub repository ↗

CONCEPT: ACCESSIBLE ROUTES, REPORTED BY THE CITY

1. Volunteers and verified users report obstacles on sidewalks and roads.
2. Each obstacle is rated by how much it blocks an accessible route.
3. People with disabilities plan walks around the worst obstacles.

GuideMe

SEPT 2026 – PRESENT CO-INITIATOR

FIXCLUJ CIVIC CAMPAIGN

A civic initiative to make walking in Cluj-Napoca easier for people with disabilities, built on crowdsourced reports of urban obstacles.

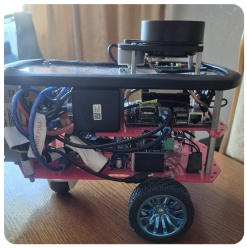
MY ROLE

Co-designed the crowdsourcing concept and helped build the online community that gathers volunteers and early feedback. The technical development is done by other team members.

Instagram ↗ LinkedIn ↗

BEYOND THE SCREEN

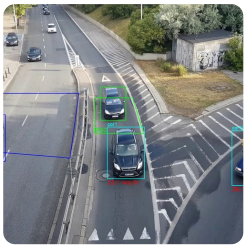
Other projects



Autonomous Mobile Robot

Bachelor's thesis. I built a robot on a Jetson Nano and an Arduino, then compared 6 SLAM mapping setups with and without sensor fusion across 120 runs.

GitHub repository ↗



Traffic Analysis with YOLO

Computer vision for traffic video: vehicle detection, licence plate reading and speed estimation, presented in a Streamlit dashboard.

GitHub repository ↗

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